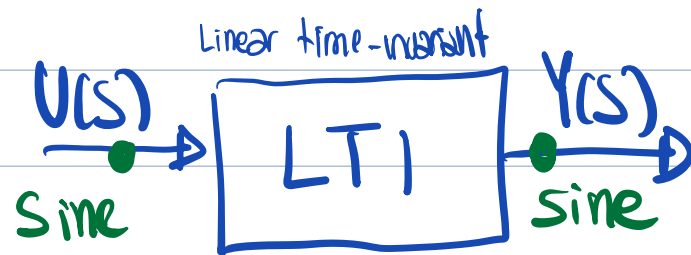


Frequency Response :

is defined as the steady state response of the system to a sinusoidal input signal



+ same frequency (ω)
+ different Amplitude (A)
+ different phase ϕ

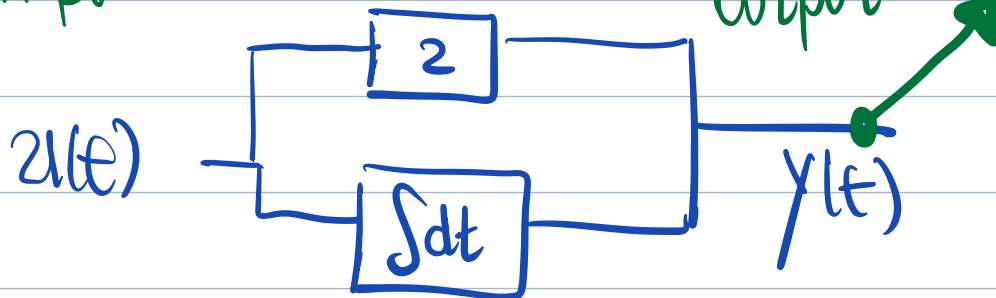
when transient response has died out.

Example

Input = $\sin(0.5t)$
 $\omega = 0.5 \text{ rad/s}$

$$y(t) = 2\sin(t \cdot 0.5) - 2\cos(0.5t)$$

input



By Trig identity
 $a \sin x + b \cos x = \sqrt{a^2 + b^2} \sin(x + \phi)$
where $\phi = \text{atan}\left(\frac{b}{a}\right)$ if $a \geq 0$

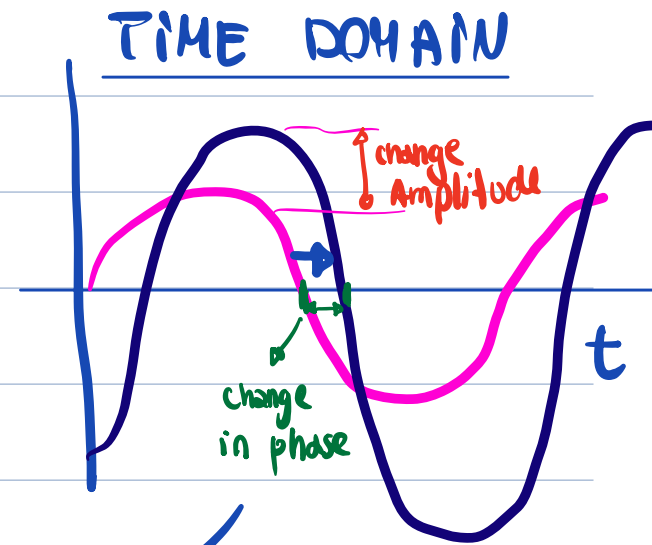
Sol) $\sqrt{8} \sin\left(\frac{1}{2}t + \text{atan}(-1)\right)$

$\approx 2.83 \sin\left(\frac{1}{2}t - 0.785\right)$

change in amplitude

frequency =

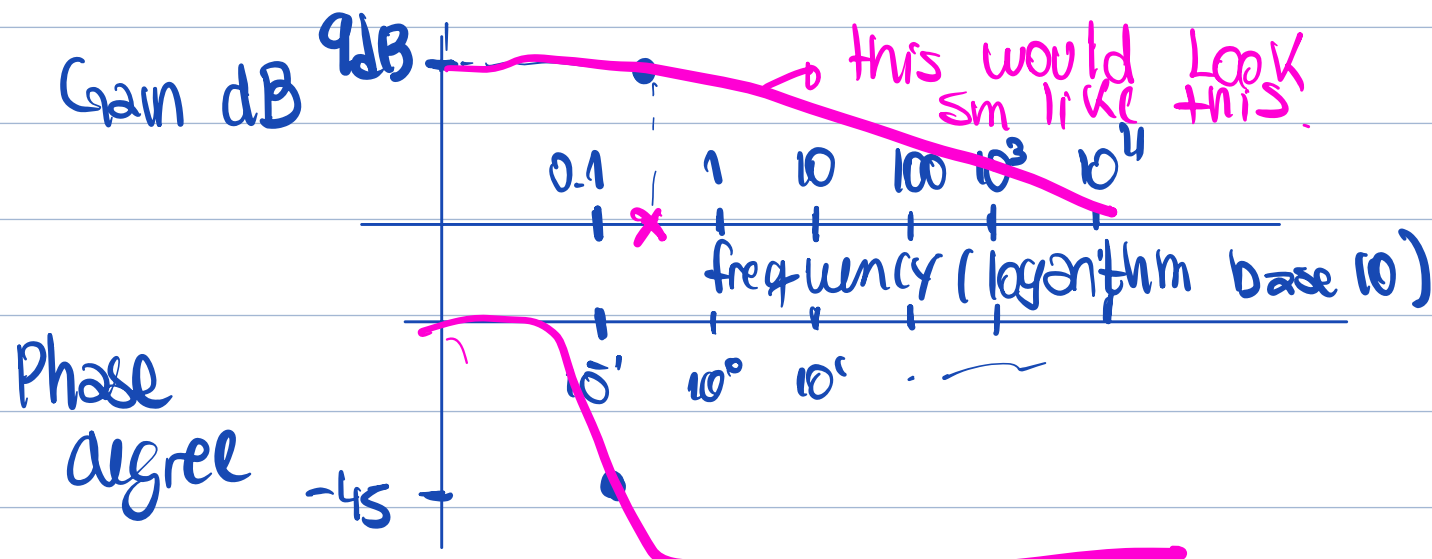
change in phase



Response just for a single freq.

What if we want to check the phase and gain in the whole spectrum (not just for $\omega = \frac{1}{2}$) ???

R// we use the Bode Plot



Frequency response information to be display graphically

we have to convert the Amplitude to decibel

From example

$$\rightarrow 20 \log_{10}(A)$$

$$\text{Gain dB} = 20 \log(2.83) \approx 9 \text{ dB}$$

$$\text{phase} = -0.785 \cdot \frac{180}{\pi} = -45^\circ$$

Bode plots rule

1) We take the transfer function $H(s)$ and replace ss freq response $H(j\omega)$ (σ has died out).

$$H(j\omega) = a + b j$$

$$\text{Gain} = |H(j\omega)| = \sqrt{a^2 + b^2}$$
$$\text{Phase. ang} \left(\frac{b}{a} \right)$$

Phase
Angles

1) $a > 0, b \geq 0$
 $\theta = \arctan\left(\frac{b}{a}\right)$

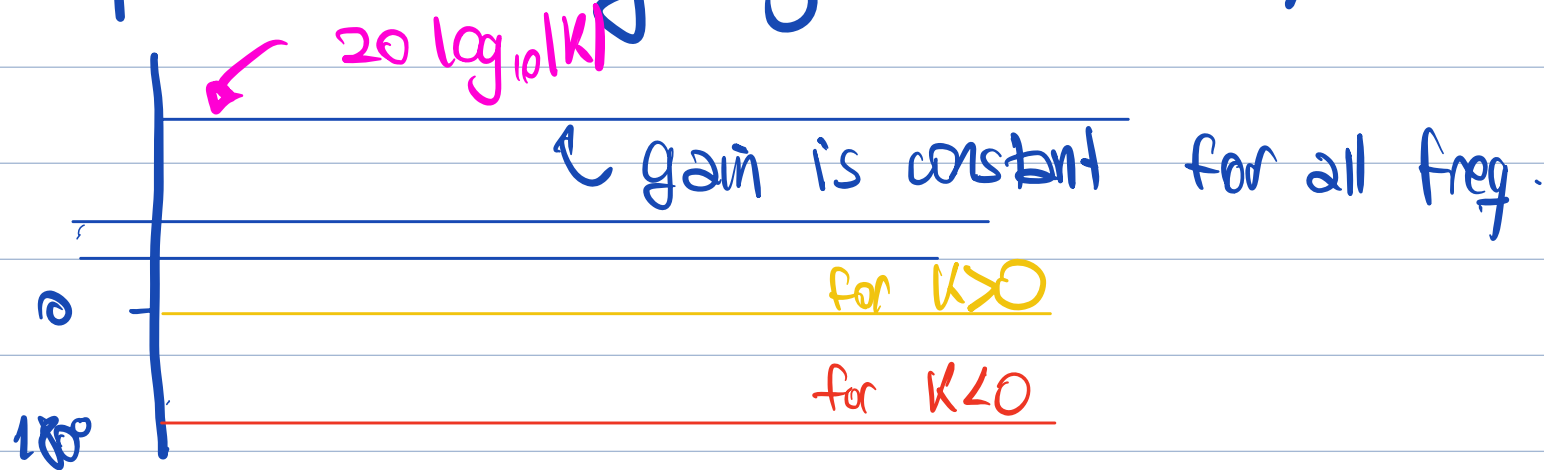
2) $a < 0, b > 0$
 $\theta = \arctan\left(\frac{b}{a}\right) + \pi$

3) $a < 0, b < 0$
 $\theta = \arctan\left(\frac{b}{a}\right) - \pi$

4) $a > 0, b < 0$
 $\theta = \arctan\left(\frac{b}{a}\right)$

2) Evaluate how each term affect the whole response.

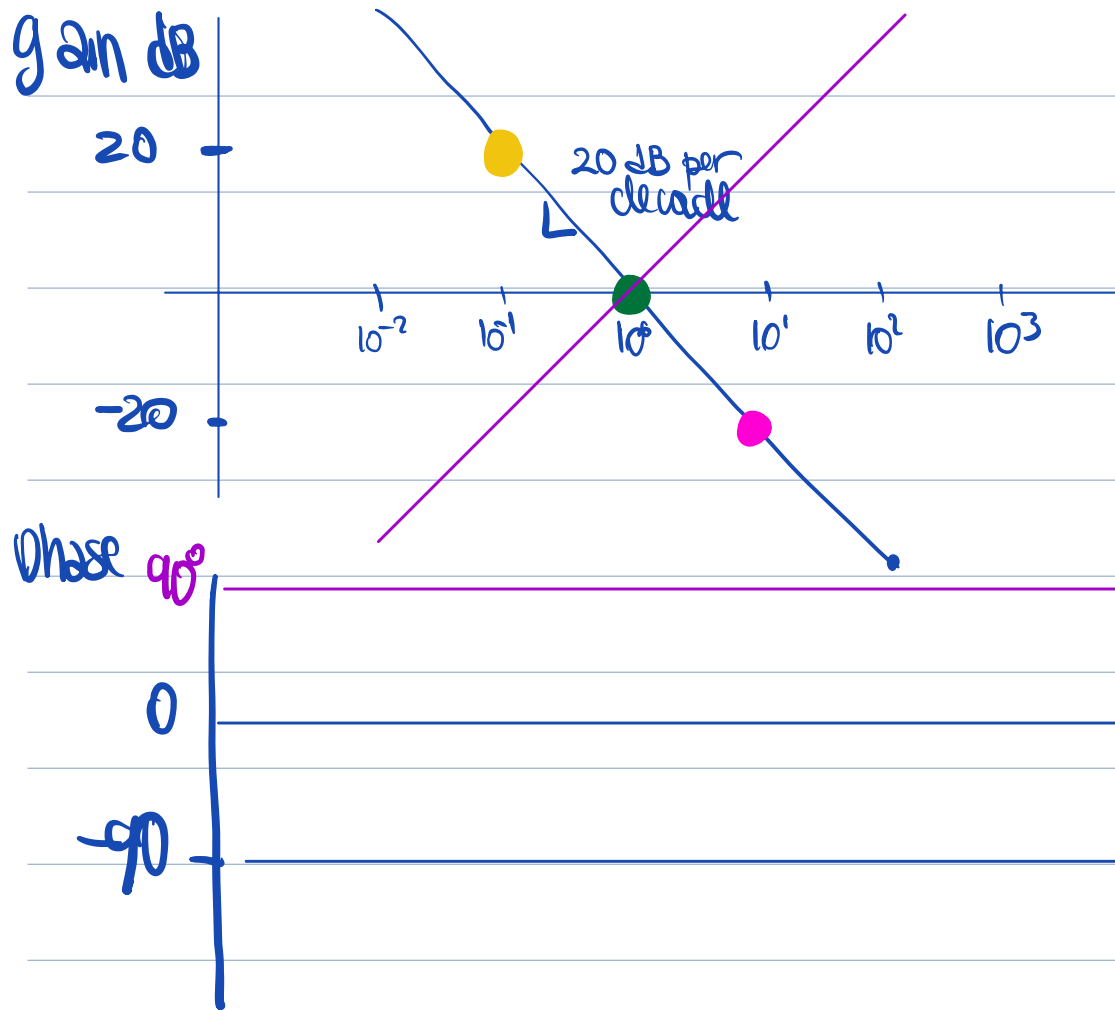
2.1) Constant $H(s) = K$ dB
gain $= |H(j\omega)| = |K| \Rightarrow 20 \log_{10}(|K|)$
phase $= \angle H(j\omega) = \angle K = \begin{cases} 0 & \text{if } K > 0 \\ \pi & \text{if } K < 0 \end{cases}$



2.2) Poles and zeros at the origin

Pole $H(s) = \frac{1}{s} \Rightarrow \frac{1}{j\omega} = -\frac{j}{\omega}$

gain $= |H(j\omega)| = \frac{1}{\omega} \xRightarrow{\text{dB}} 20 \log_{10} \left(\frac{1}{\omega} \right) = -20 \log_{10}(\omega)$
phase $= \angle \left(-\frac{j}{\omega} \right) = -\frac{\pi}{2}$



- when $\omega = 1$
 $20 \log(1) = 0$ •
- when $\omega = 10$
 $20 \log\left(\frac{1}{10}\right) = -20$ •
- when $\omega = 0.1$
 $20 \log(10) = 20$ •

For a zero (plot in magenta).

2.3) Real poles or zeros

$$H(s) = \frac{1}{1 + \frac{s}{\omega_0}}$$

$$\xrightarrow{s=j\omega}$$

$$\frac{1}{1 + \frac{j\omega}{\omega_0}}$$

"break frequency"

We took $H(j\omega)$ and separate the real and Im. part.

The gain in dB $\Rightarrow -20 \log_{10} \left(\sqrt{1 + \frac{\omega^2}{\omega_0^2}} \right)$

The phase equation $\Rightarrow \text{atan}\left(-\frac{\omega}{\omega_0}\right)$

Case 1) $\omega \ll \omega_0$

Gain $-20 \log_{10} \left(\sqrt{1+0} \right)$ Gain

$-20 \log_{10}(1) = 0$

Phase $= \text{atan}(0) = 0^\circ$

Case 2) $\omega = \omega_0$

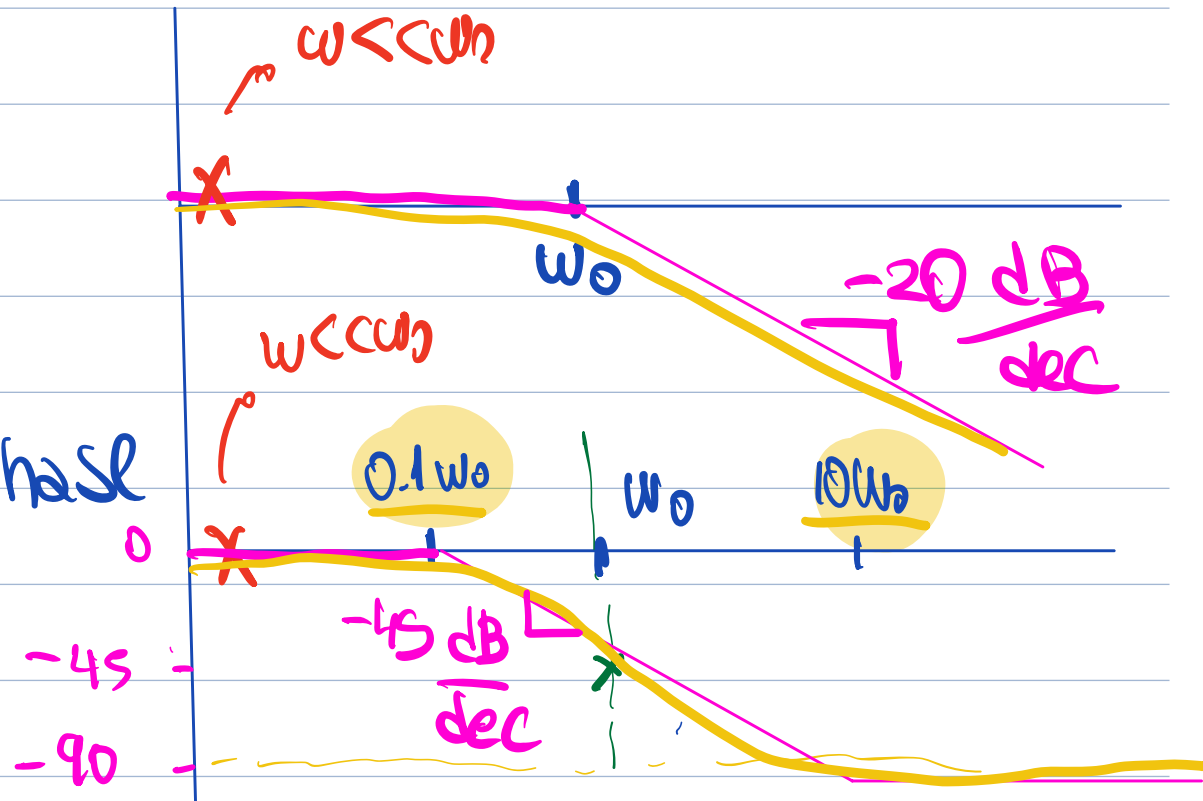
Gain $= -20 \log_{10}(\sqrt{2}) = -3 \text{ dB}$

Phase $(-1) = -45 \text{ deg}$

Case 3) $\omega \gg \omega_0$

Gain $= -20 \log_{10} \left(\frac{\omega}{\omega_0} \right) = -20 \frac{\text{dB}}{\text{dec}}$

Phase $= \text{atan}(\infty) = -90 \text{ deg}$



For phase plot we normally calculate for $0.1\omega_0$, ω_0 and $10\omega_0$

2.4) Complex poles

$$H(s) = \frac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2} = \frac{1}{\left(\frac{s}{\omega_0}\right)^2 + 2\zeta\left(\frac{\omega}{\omega_0}\right) + 1}$$

$$\text{Real} = \frac{1 - \left(\frac{\omega}{\omega_0}\right)^2}{\left(1 - \left(\frac{\omega}{\omega_0}\right)^2\right)^2 + \left(2\zeta\left(\frac{\omega}{\omega_0}\right)\right)^2}$$

$$\text{Im} = \frac{-\left(2\zeta\left(\frac{\omega}{\omega_0}\right)\right)}{\left(1 - \left(\frac{\omega}{\omega_0}\right)^2\right)^2 + \left(2\zeta\left(\frac{\omega}{\omega_0}\right)\right)^2}$$

Case 1) $\omega \ll \omega_0$

$$\text{Real} = 1 \quad \text{Im} = 0$$

$$|H(j\omega)| = 20 \log(\sqrt{1+0}) = 0 \text{ dB}$$

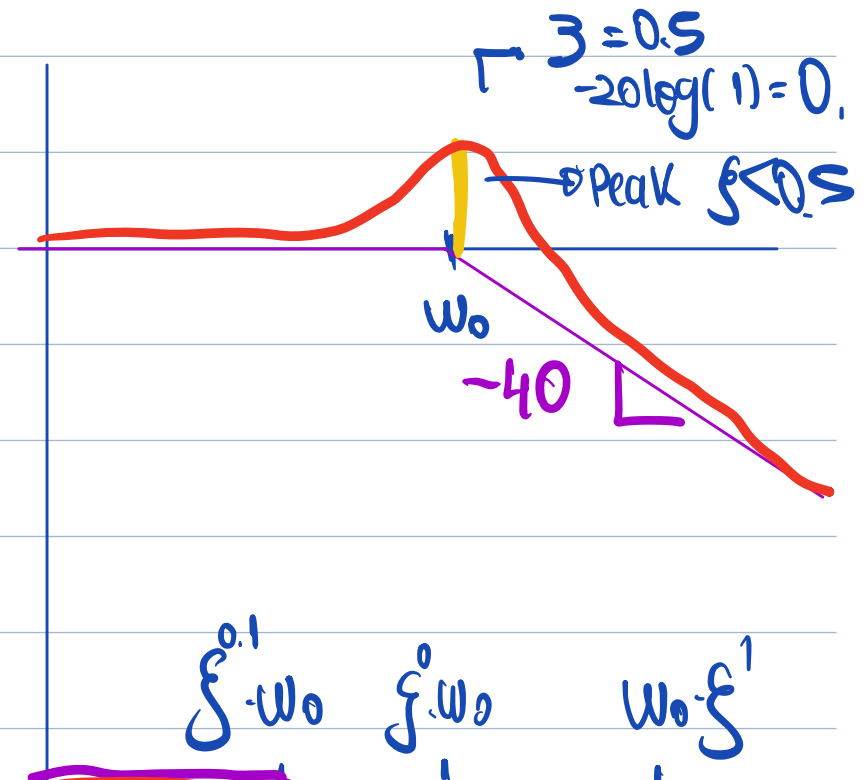
$$\text{Phase}(H(j\omega)) = \arctan\left(\frac{0}{1}\right) = 0^\circ$$

Case 2) $\omega \gg \omega_0$ $\text{Real} = -\left(\frac{\omega}{\omega_0}\right)^{-2}$ $\text{Im} = -\left(2\zeta\frac{\omega}{\omega_0}\right)^{-3}$

$$|H(j\omega)| = -40 \log\left(\frac{\omega}{\omega_0}\right) \rightarrow -40 \text{ dB/dec}$$

$$\text{Phase}(H(j\omega)) = \arctan\left(2\zeta\left(\frac{\omega}{\omega_0}\right)^{-1}\right) = -180^\circ$$

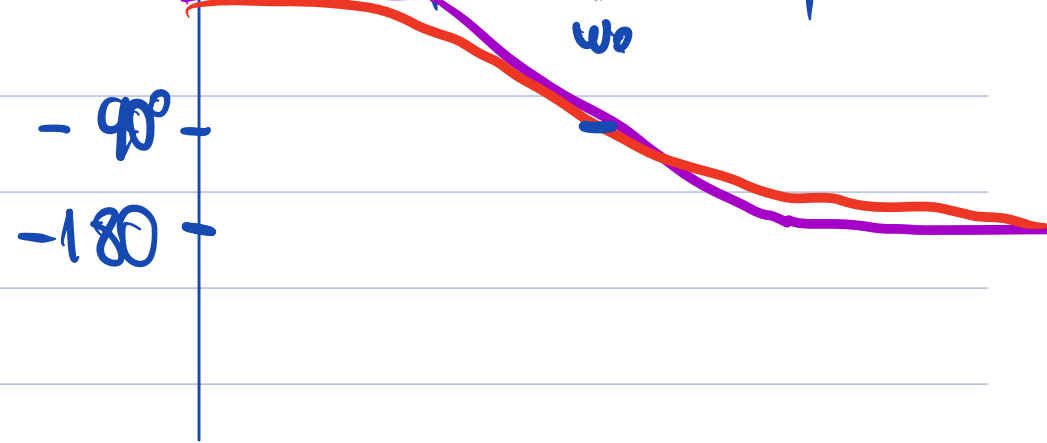
Case 3) $\omega = \omega_0$ $\text{Real} = 0$ $\text{Im} = -\frac{1}{2\zeta}$



$$|H(j\omega)| = 20 \log \left(\frac{1}{2\xi} \right) = -20 \log(2\xi)$$

$$\downarrow \xi \rightarrow \uparrow |H(j\omega)| \quad (\xi=0, |H|=\infty)$$

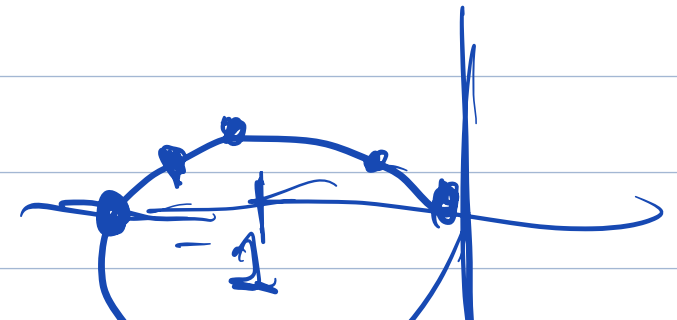
$$\arg(H(j\omega)) = \alpha \tan(\infty) = -90^\circ$$



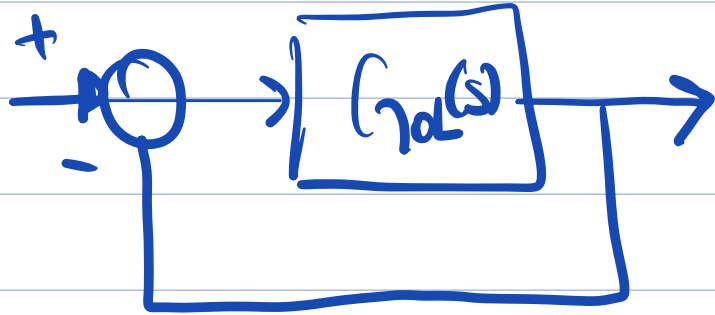
BANDWIDTH : In terms of dB, it is defined as the freq. at which $20 \log_{10} |M(j\omega)|$ does not fall below -3 dB



3) Add all the effect from each pole/zero, constant to a single plot!!



Gain & Phase Margin



$$G_{cl} = \frac{G_d}{1 + G_d}$$

Recall:

THE SYSTEM is unstable if it has a pole in the RHP

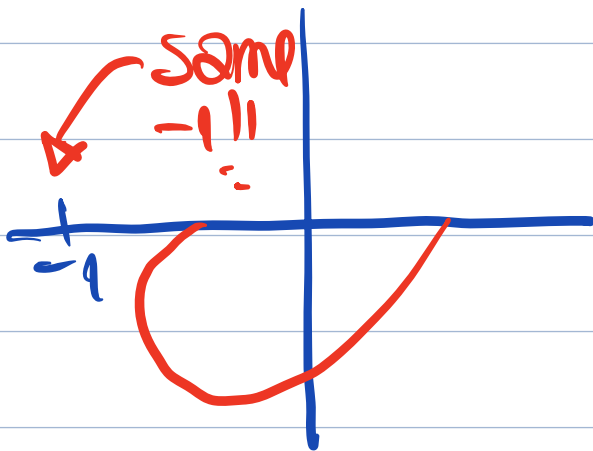
Or.. gain of the closed loop system is ∞ !!

$$\frac{G(s)}{1 + G_d} = \infty \implies \text{or } 1 + G(s) = 0 \implies G(s) = -1$$

we avoid this
... Look similar to?

Recall

NYQUIST !!



-1

- Gain = $|-1| = 1 \Rightarrow 0 \text{ dB}$
- Phase = -180°

If any freq. produces 0dB gain AND 180° phase, the closed loop system is unstable

Let's return to Bode

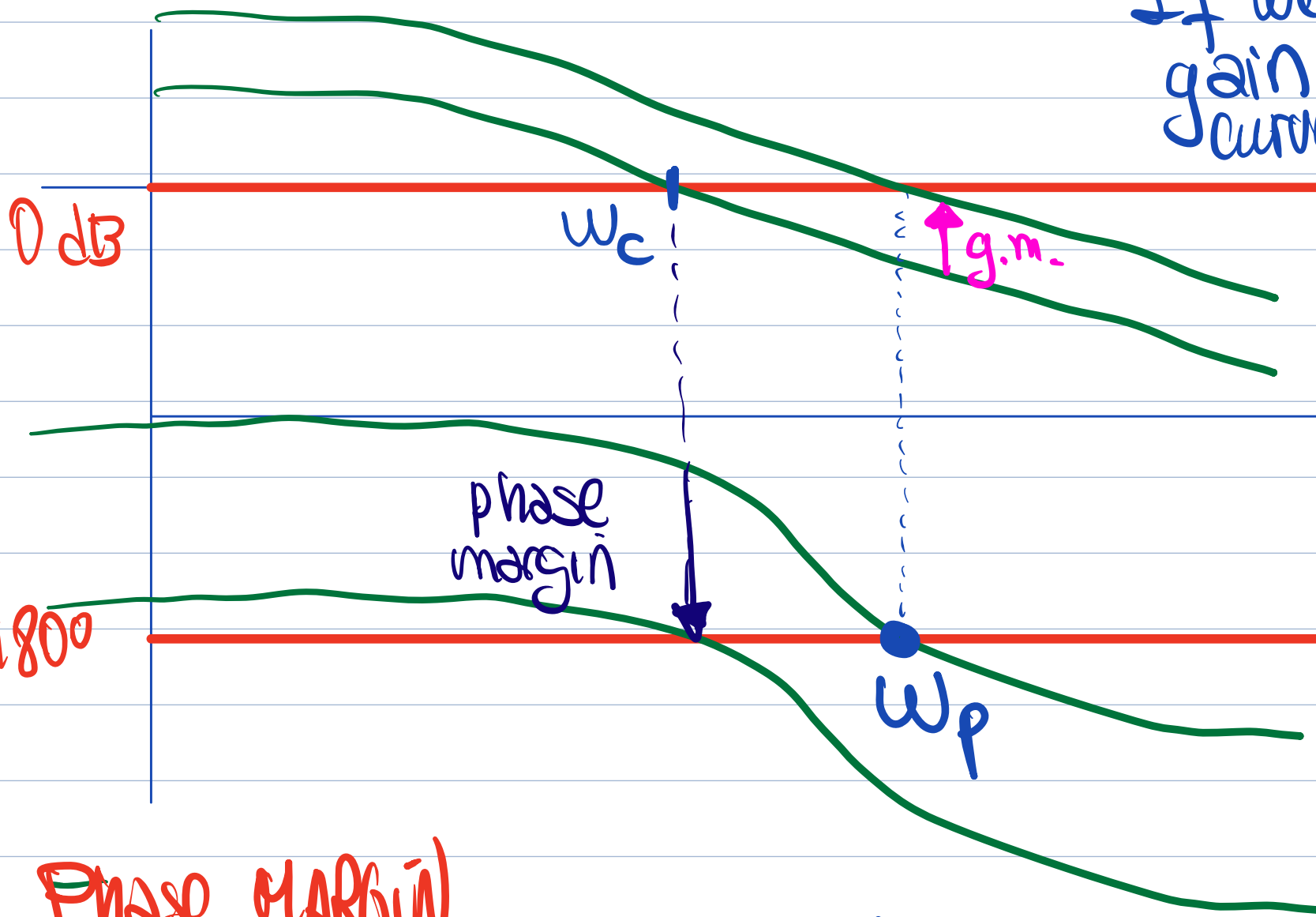
Gain MARGIN

If we increase the gain, the magnitude curve, move up.

Gain margin so we have

-1

this is the K.
does not affect the phase plot



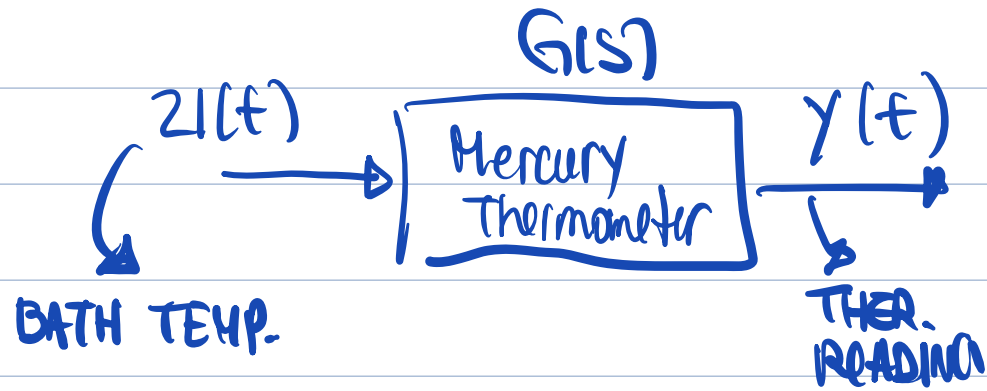
Phase MARGIN

How much phase lag would it take to make -180° at the 0° dB gain

Add a poles tends to delay the phase
add a zero tends to advance the phase

4.1

poles reduce phase margin
zero increase phase margin



$$G(s) = \frac{k}{(s + a)} \quad \text{First order system.}$$

Input $u(t)$

$$\text{Magnitude} = \frac{\text{Max} - \text{Min}}{2} = \frac{32 - 28}{2} = 2$$

$$\text{frequency} = \frac{2\pi}{0.314 \text{ min} \cdot \frac{60 \text{ [sec]}}{1 \text{ [min]}}} = 0.052\pi = 0.33 \left[\frac{\text{rad}}{\text{s}} \right]$$

Output

$$\text{Magnitude} = \frac{\text{Max} - \text{Min}}{2} = \frac{30.9 - 29.1}{2} = \frac{1.8}{2} = 0.9$$

$$\phi_{\text{shift}} = \omega \cdot \Delta t = \frac{2\pi}{60 \cdot 0.314} \cdot 60 \cdot 0.056 = -1.12 \text{ rad}$$

From freq. response we know

$$u(t) = A \sin(\omega t + \theta) \rightarrow \boxed{G(s)} \xrightarrow{y(t)} y(t) = A |G(j\omega)| \sin(\omega t + \theta + \angle G(j\omega))$$

From graph $\theta = 0^\circ$

$$\begin{aligned} \text{Eq that we have : } A \cdot |G(j\omega)| &= 2 \cdot |G(j\omega)| = 0.9 \\ |G(j\omega)| &= 0.45 \end{aligned}$$

$$G(s) \Rightarrow G(j\omega) = \frac{K}{j\omega + \alpha}$$

$$|G(j\omega)| = \frac{K}{\sqrt{\omega^2 + \alpha^2}}$$

$$\angle G(j\omega) = 0 - \tan^{-1}\left(\frac{\omega}{\alpha}\right) = -\tan^{-1}\left(\frac{\omega}{\alpha}\right)$$

$$(1) K^2 = 0.45^2 (\omega^2 + \alpha^2) = 0.45^2 (0.33^2 + \alpha^2)$$

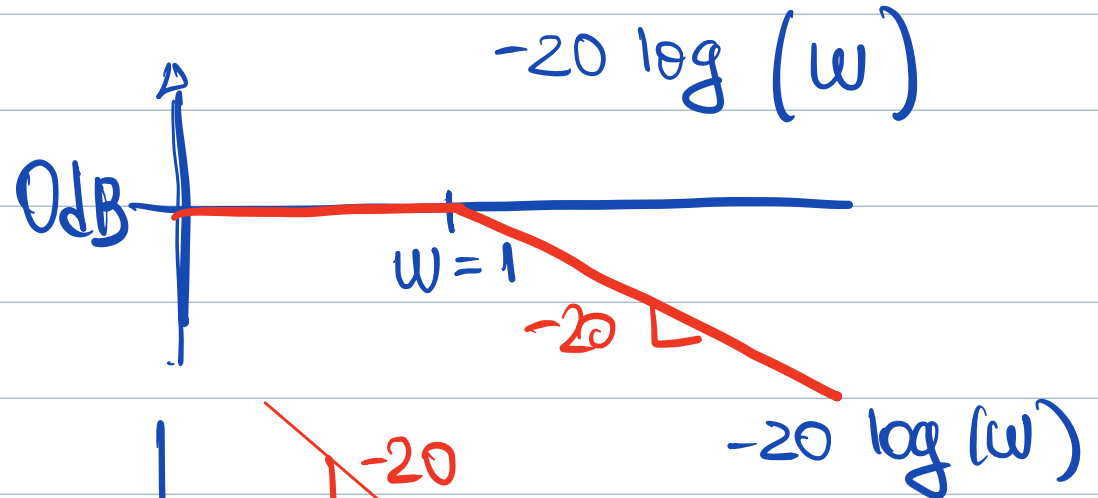
$$(2) \quad \tan(-1.12) = -\frac{\omega}{\alpha} \Rightarrow 2.066 = \frac{0.33}{\alpha}$$

$$\alpha = 0.16 //$$

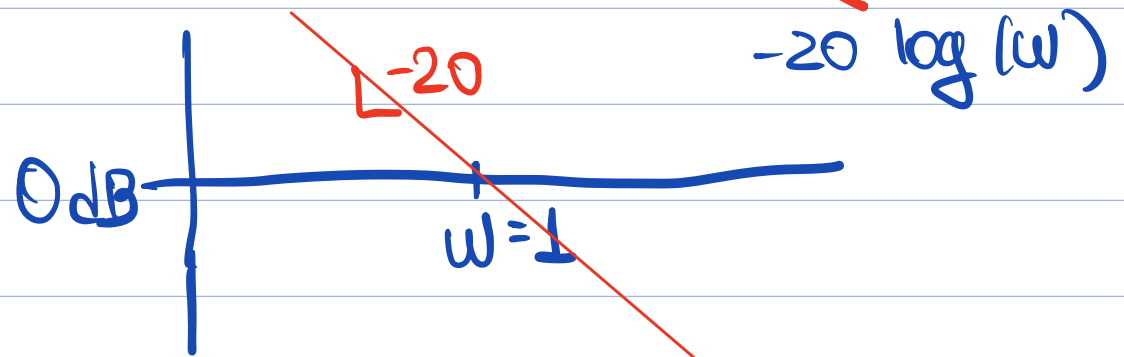
$$K = \pm 0.16 \Rightarrow K = 0.16 //$$

4.9) Combine the plots

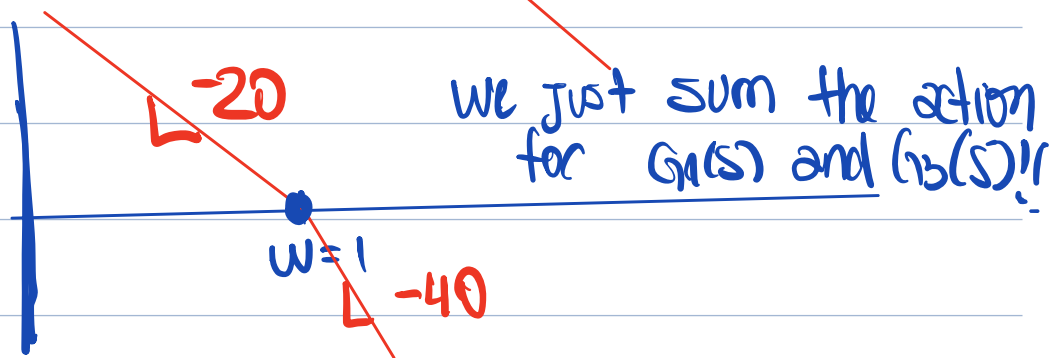
$$G_1(s) \Rightarrow \frac{1}{j\omega + 1} \approx \frac{1}{j(\frac{\omega}{\omega_0}) + 1} \quad \therefore \omega_0 = 1$$



$$G_3(s) \Rightarrow \frac{1}{j\omega}$$



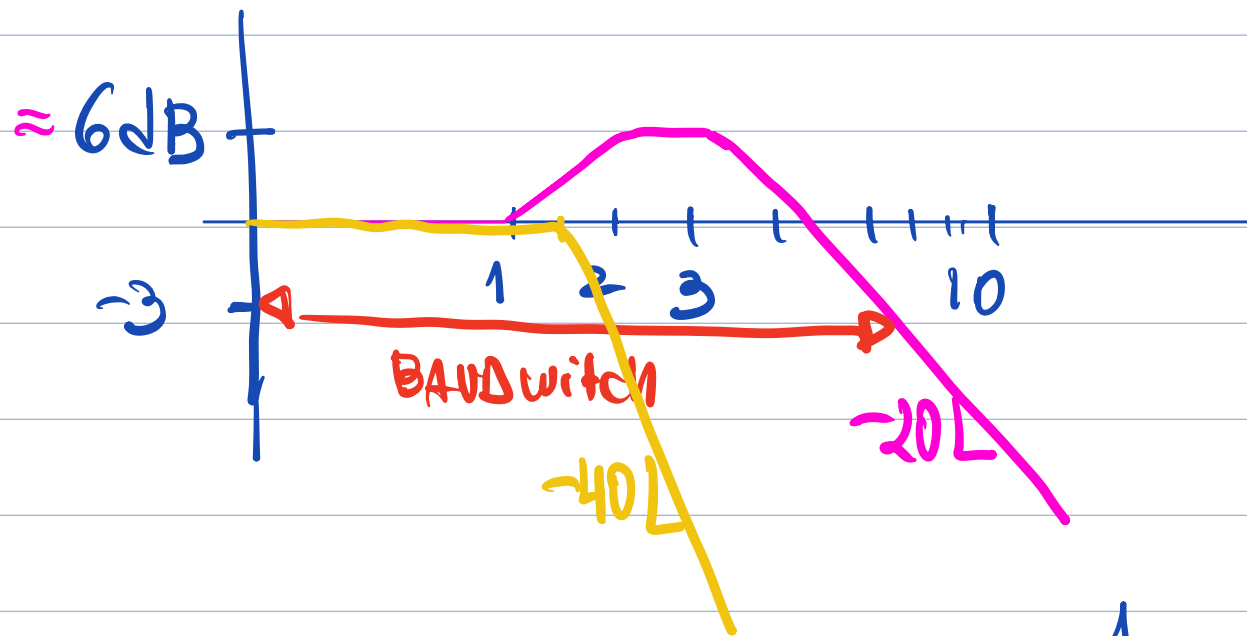
$$G_4(s) \Rightarrow \frac{1}{j\omega(j\omega + 1)}$$



$$G_2(s) = \frac{6(s+1)}{(s+2)(s+3)} = \frac{(j\omega+1)}{\left(\frac{j\omega}{2}+1\right)\left(\frac{j\omega}{3}+1\right)}$$

Draw in pink

$$20 \log\left(\frac{\omega}{\omega_0}\right) = 6 \text{ dB}$$



$$G_5(s) = \frac{5}{s^2 + 2s + 5} = \frac{1}{\left(\frac{s}{\sqrt{5}}\right)^2 + \frac{2}{5}s + 1}$$

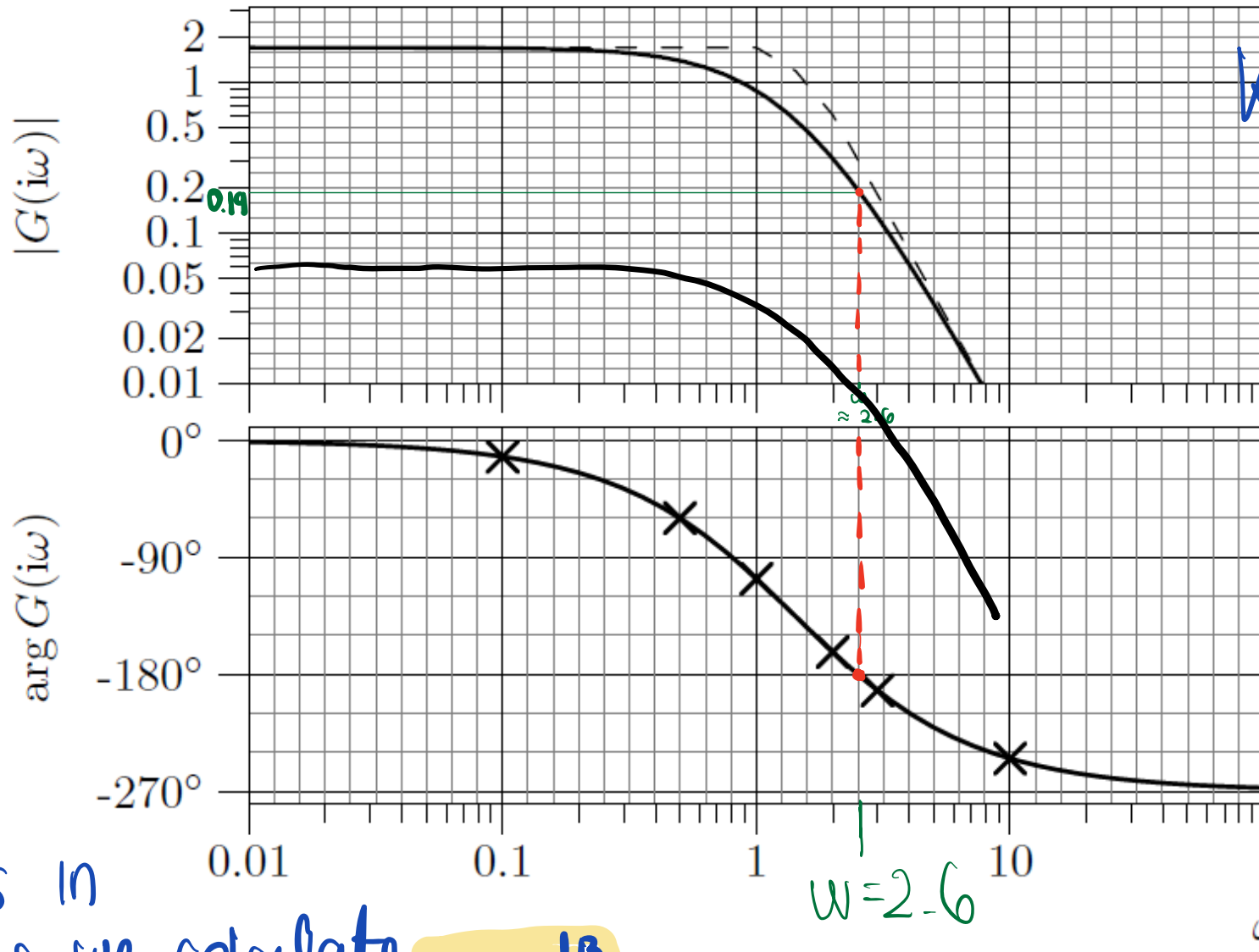
$$\omega_0 = \sqrt{5} \approx 1.58$$

DRAW in yellow

$\therefore$ BANDWIDTH of $G_2(s) > B.$ of $G_5(s)$

We need to scale the gain plot to make the gain in $\omega = 2.6$ to be at 0dB $\Rightarrow$ which is the gain 0.19

Amplitude $\uparrow$



If it's in decibels we calculate $20 \log(K) = \text{dB}$

$$K = 10^{\frac{\text{dB}}{20}}$$

Figure 4.11a

4 Frequency Description

4.1 A mercury thermometer can be described with high accuracy as a first order linear time invariant dynamic system. The input is the real temperature and the output is the thermometer reading. In order to decide the transfer function in a thermometer it is placed in liquid where the temperature is varied as a sinusoid. The obtained result is shown in Figure 4.1a. Find the transfer function of the thermometer.

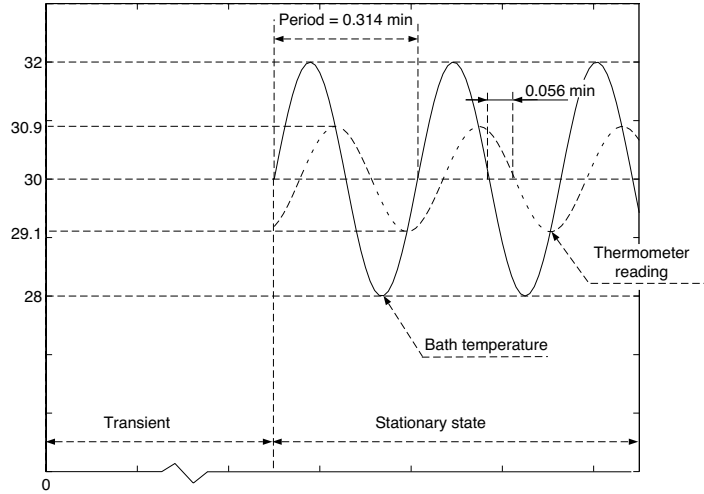


Figure 4.1a

4.2 We want to keep a ship on a given course, Ψ , with an automatic control system using the rudder angle δ . See Figure 4.2a. If ω denotes the angular velocity of the ship,

$$\omega = \dot{\Psi} \quad (4.1)$$

the following differential equation is valid for small values of ω and δ ,

$$T_1 \dot{\omega} = -\omega + K_1 \delta \quad (4.2)$$

where $T_1 = 100$ and $K_1 = 0.1$. The desired course, Ψ_{ref} , and the measured course, Ψ , are fed in to the auto pilot, which gives the signal u to the rudder

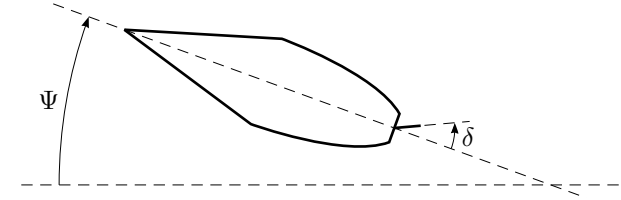


Figure 4.2a

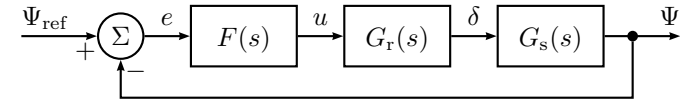


Figure 4.2b

engine. Figure 4.2b shows a block diagram of the auto pilot. The auto pilot has the transfer function

$$F(s) = K \frac{1 + \frac{s}{a}}{1 + \frac{s}{b}}, \quad a = 0.02, b = 0.05$$

while G_r is given by

$$G_r(s) = \frac{1}{1 + sT_2}, \quad T_2 = 10$$

and $G_s(s)$ is defined by (4.1) and (4.2).

a) Make a Bode plot for the transfer function FG_rG_s , for $K = 0.5$.

4. Transfer function $G_S(s)$

$$T_1 \ddot{\psi} = -\dot{\psi} + K_1 \delta$$



$$T_1 s^2 \psi(s) = -s \psi + K_1 \delta(s)$$

$$\frac{\psi(s)}{\delta(s)} = \frac{K_1}{s(T_1 s + 1)}$$

Bode plot of

$$F \cdot G_r \cdot G_S = \frac{K_1}{s(T_1 s + 1)} \cdot \frac{1}{1 + s T_2} \cdot \frac{K \left(1 + \frac{s}{a}\right)}{1 + \frac{s}{b}}$$

$$= \frac{0.1 \cdot 0.5}{s(100s + 1)} \cdot \frac{1}{(1 + s \cdot 10)} \cdot \frac{\left(1 + \frac{s}{0.02}\right)}{\left(1 + \frac{s}{0.05}\right)}$$

$$G(j\omega) = \frac{0.05 \left(1 + \frac{s}{0.02}\right)}{s \left(\frac{s}{0.01} + 1\right) \left(\frac{s}{0.1} + 1\right) \cdot \left(\frac{s}{0.05} + 1\right)}$$

① For low frequencies $|G(j\omega)| = \frac{0.05}{\omega}$

$\rightarrow \omega = 0.001 \Rightarrow |G(j\omega)| = 50$ then $20 \log(50) = 33$ dB

$$\angle \tan \left(-\frac{1}{\omega}, 0 \right) = -90^\circ$$

② Rest of points

$$|G(j\omega)| = 0.05 \sqrt{1 + \left(\frac{\omega}{0.02}\right)^2}$$

$$\frac{\omega \sqrt{1 + \left(\frac{\omega}{0.01}\right)^2} \sqrt{4 + \left(\frac{\omega}{0.05}\right)^2} \sqrt{1 + \left(\frac{\omega}{0.1}\right)^2}}{}$$

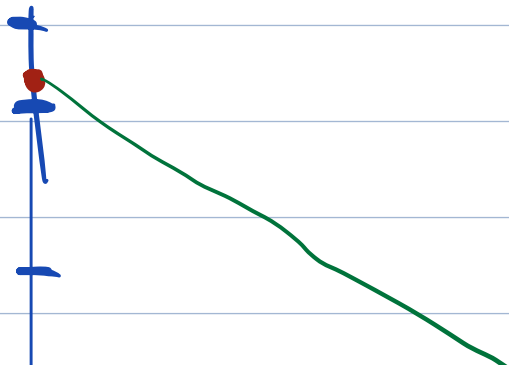
$$\begin{aligned} \angle G(j\omega) = & \arctan\left(\frac{\omega}{0.02}\right) - 90 - \arctan\left(\frac{\omega}{0.01}\right) \\ & - \arctan\left(\frac{\omega}{0.05}\right) - \arctan\left(\frac{\omega}{0.1}\right) \end{aligned}$$

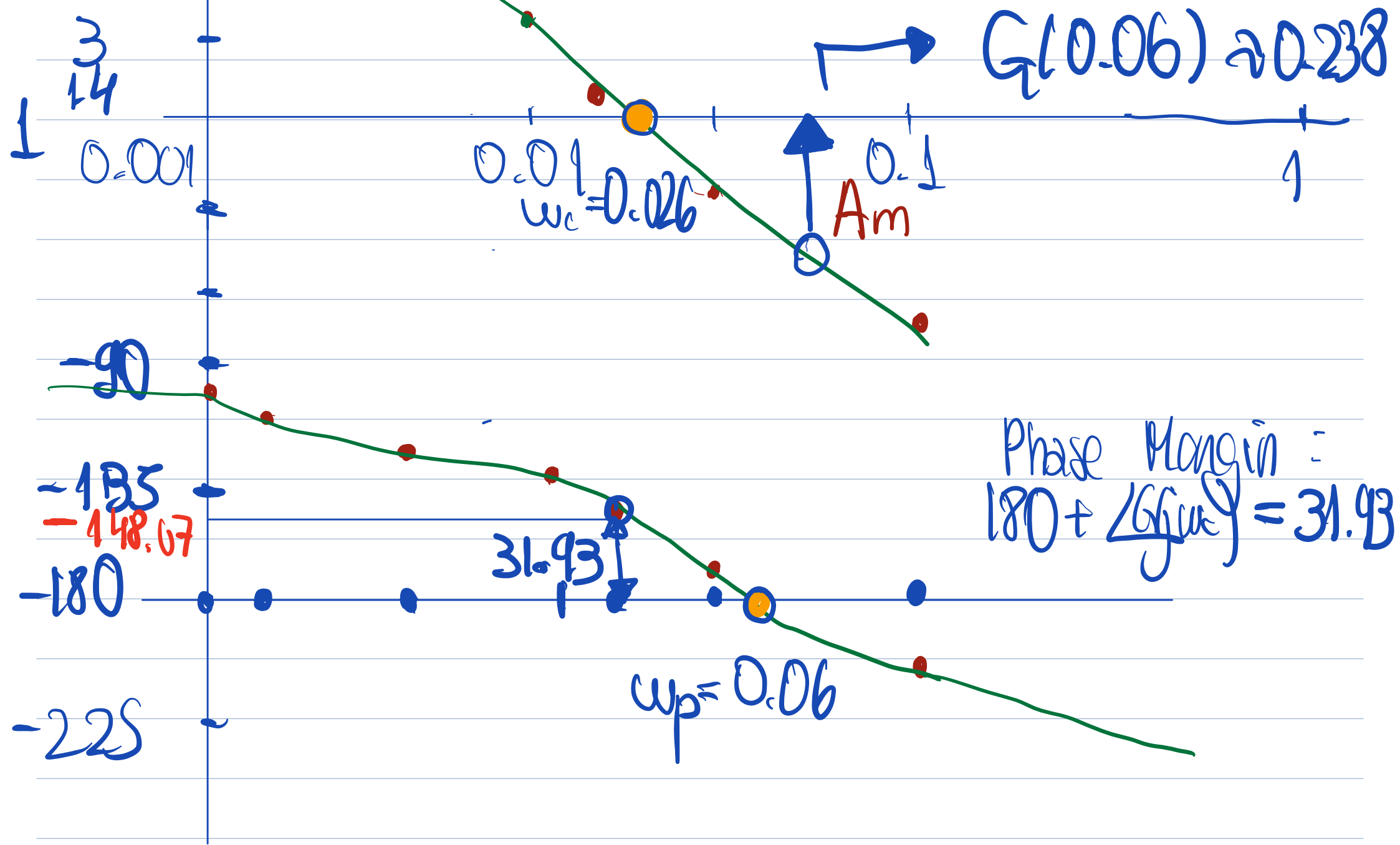
$$50 \quad 3.8568 \quad 1.4395 \quad 0.333 \quad 0.08$$

		0.01		0.02		0.05		0.1	
Slope	-20	11.72	-40	3.16	-20	-9.52	-40	-21.94	-60

PHASE		0.001		0.002		0.005		0.01	
		-94	-45s	-99	0s	-111	-45s	-125	-90
		0.02		0.05		0.1		0.2	
		-141	-90	-172	-90	-204	-45	-232	-90
		0.5		1					
			-45		0				

50





What is A_m

$$|G(0.06)| \approx 0.238$$

Then if I multiply by $\frac{1}{0.238}$
at $\omega_p = 0.06$ I will make the gain
to be 1. Then

$$A_m = 4.2$$

b) We calculated this before

For making the system to start

oscillating $\hookrightarrow$ we need to check gain margin
we calculated $A_m = 4.2$

Then we need to compute the K_{crit} as

$$K_{crit} = 0.5 \cdot A_m = 2.1$$

which oscillates as $\omega_p = 0.06$.

$$c) \quad \psi_{ref} = 5 \sin(0.02t)$$

$$\psi_{ss}(t) = B \sin(\beta t + \varphi)$$

with $K=5$ what do we have

Since this is a LTI, the frequency remains the same, then

$$\beta = \alpha = 0.02$$

we need to check the value of the magnitude and phase for $\omega = 0.02$

CAN I USE THE PLOT ??

NO... that was open loop

But we can use

$$G_{OL} = G(S) \Rightarrow G_{CL} = \frac{G_{OL}}{1 + G_{OL}}$$

Then we compute gain and phase for $G_{OL}(j\omega)$

$$|G_{OL}(0.02j)| = 1.44$$

$$\angle G_{OL}(0.02j) = -142^\circ$$

$$-1.135 - j0.886$$

GAIN $\approx$

Then

$$G_{CL} = \frac{1.44}{\sqrt{1.135^2 + 0.886^2}} = 1.61$$

$$\text{Then } B = A \cdot |G_{CL}| = 5 \cdot 1.61 = 8$$

PHASE :

$$\begin{aligned}\angle G_{cl}(0.02j) &= -142 - \angle(1 + G(0.02j)) \\ &= -142 - \angle -0.135 - j0.886 \\ &= -142 - \text{atan}\left(\frac{0.886}{0.135}\right) \approx -133^\circ \\ &\approx \text{rad} = -0.76\end{aligned}$$

$$\therefore \beta = 0.02, \quad B = 8, \quad \varphi = -0.76$$

5.8 a)

The delay has magnitude $|e^{j\omega T}| = 1$

But phase equal to $-\omega T$

$\therefore$ Magnitude does not change

By looking at $|G| = 1$ the magnitude is for $\omega = 1$ with angle -140° .

$$-140^\circ \xrightarrow{\text{rad}} PM = -140 + 180 = 40^\circ \approx 0.69^\circ$$

The system is stable if $PM > 0$

with delay

$$PM - \omega T > 0$$

$$0.698^\circ - (1) \cdot T > 0$$

$$T < 0.698[\text{s}]$$

Remember $e^{-j\omega T} = \cos(\omega T) - j\sin(\omega T)$

$$|e^{-j\omega T}| = 1$$

$$\angle e^{-j\omega T} = -\omega T$$

4.2a.

From diagram

$$\omega_c = 0.079$$

$$\omega_p = 0.28$$

Magnitude at ω_p is 0.2

Then $GM = 5$

Angle at ω_c is -94.87

then $PM = 88.13^\circ$

For $PM = 50^\circ \Rightarrow L = -130^\circ$

this happens at $\omega = 0.15 \text{ rad/s}$

For $K = \frac{1}{0.525} = 1.9$ we get

Magnitude 1 and $PM = 50^\circ$

If we use $\omega^* > 0.15$ then

$PM_{\omega^*} < 50^\circ$

- b) At the testing of the auto pilot we do the following experiment. The gain of the auto pilot K is increased until the control system oscillates with constant amplitude. At what value of K does this occur? What is the period time of the oscillation?
- c) Ψ_{ref} is allowed to vary as a sinusoid

$$\Psi_{\text{ref}}(t) = A \sin \alpha t$$

where $A = 5^\circ$ and $\alpha = 0.02$. When the movements of the ship have stabilized we have

$$\Psi(t) = B \sin(\beta t + \varphi)$$

What values do B , β , and φ have if $K = 0.5$?

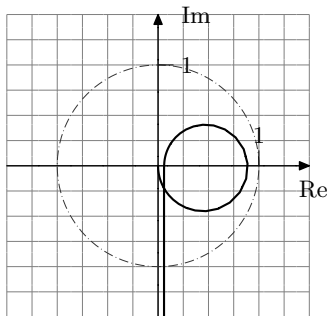


Figure 4.3a

- 4.3 a) In Figure 4.3a the Nyquist curve for a system is shown. Draw the Bode plot for the same system. The scale on the ω -axis is not important, as long as the amplitude and phase curve are in agreement.
- b) Draw a diagram for the poles and zeros for the system. The relative placement is important, not the scale.
- 4.4 Figure 4.4a shows the step responses (when the input is a unit step) and Bode gain plots of four different systems, in no particular order. Identify the pair of plots that belongs to each system. That is, for each step response, find the corresponding Bode gain plot (amplitude curve). Motivate your answer by pointing out a set of unique features for each system.

i 10

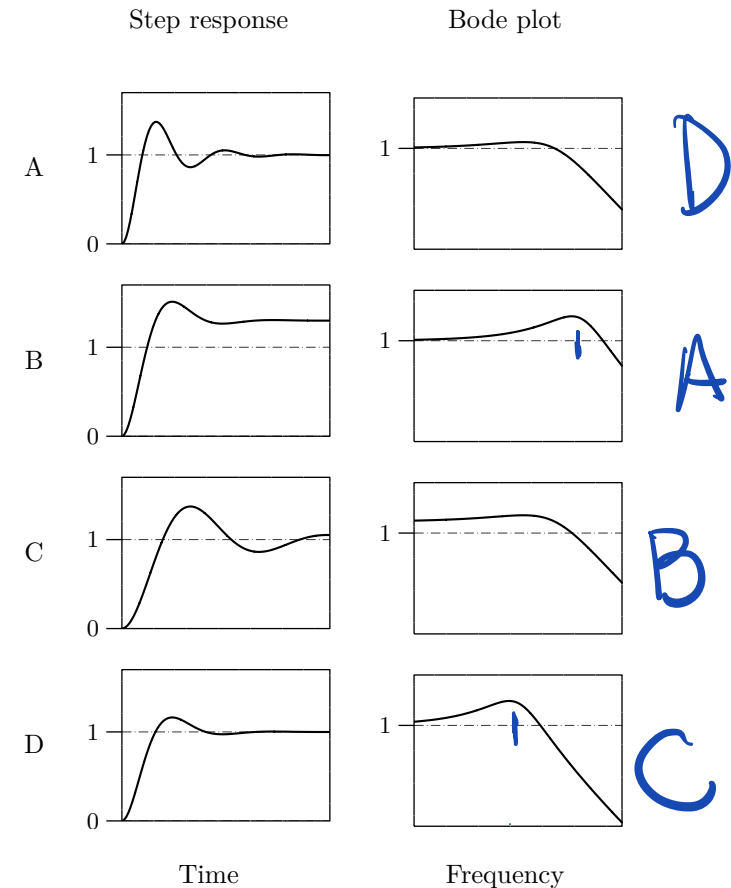


Figure 4.4a. All comparable diagrams have equal scaling.

$$\omega = 2\pi \cdot f =$$

$$\omega f \Rightarrow f \uparrow$$

They give $|G(j\omega)|$ in y-axis (gain plot)

1) Final value theorem

$$\lim_{s \rightarrow 0} s Y(s) = \lim_{s \rightarrow 0} s \cdot G(s) \Rightarrow \lim_{j\omega \rightarrow 0} G(j\omega) = G(\omega=0)$$

Plot B has an offset then $G(0) \neq 1 \Rightarrow$ Plot 3
So B-3

2) A and C presents oscillations (there is a second order system involved). Second order have resonance peak frequencies which are seen in 2 and 4. $\Rightarrow \omega_2 > \omega_4$
 ω_2 is faster, then A-2 and C-4

3) zero + poles (goes up a little, then down)

- 4.5 a) Consider the transfer functions $G_A(s)$, $G_B(s)$, $G_C(s)$, $G_D(s)$, and $G_E(s)$ in Problem 2.6

$$\begin{aligned} G_A(s) &= \frac{1}{s^2 + 2s + 1}, & G_B(s) &= \frac{1}{s^2 + 0.4s + 1} \\ G_C(s) &= \frac{1}{s^2 + 5s + 1}, & G_D(s) &= \frac{1}{s^2 + s + 1} \\ G_E(s) &= \frac{4}{s^2 + 2s + 4} \end{aligned}$$

Study the amplitude curves of the Bode plots for the systems and find the static gain and bandwidth of the systems. In cases when it is relevant find also the resonance frequency and resonance peak.

- b) Describe qualitatively (without formulas) the relationships between T_r (rise time) and ω_B (bandwidth) and between M (overshoot) and M_p (resonance peak) respectively?

- 4.6 A system has the transfer function

$$G(s) = \frac{e^{-2s}}{s(s+1)}$$

What is the output (after transients) when the input is

$$u(t) = 2 \sin(2t - 1/2)$$

- 4.7 For the systems below the input is chosen as $u(t) = \sin(2t)$. Determine the output signal $y(t)$ after transients have faded away, provided that it exists.

- a) $Y(s) = \frac{1}{s+1}U(s)$
b) $Y(s) = \frac{1}{s-1}U(s)$
c) $Y(s) = \frac{1}{(s+1)(2s+1)}U(s)$
d) $Y(s) = \frac{e^{-0.5s}}{s+1}U(s)$

- 4.8 A system is described by $Y(s) = G(s)U(s)$. Figure 4.8a shows $u(t) = \sin(\omega t)$ and the corresponding output $y(t)$ (after all transients have faded away) for the frequencies $\omega = 1, 5, 10$, and 20 rad/s (from top to bottom).

- a) Determine the gain ($|G(i\omega)|$) and phase ($\arg G(i\omega)$) for the system for each value of ω .
b) Determine the gain values in dB₂₀ ($20 \log_{10}(|G(i\omega)|)$).
c) Sketch the Bode plot using the values determined above.

- 4.9 Combine the transfer functions below with the Bode plots in Figure 4.9a

$$\begin{aligned} G_1(s) &= \frac{1}{s+1}, & G_2(s) &= \frac{6(s+1)}{(s+2)(s+3)} \\ G_3(s) &= \frac{1}{s}, & G_4(s) &= \frac{1}{s(s+1)} \\ G_5(s) &= \frac{5}{s^2 + 2s + 5} \text{ (poles: } -1 \pm i2) \end{aligned}$$

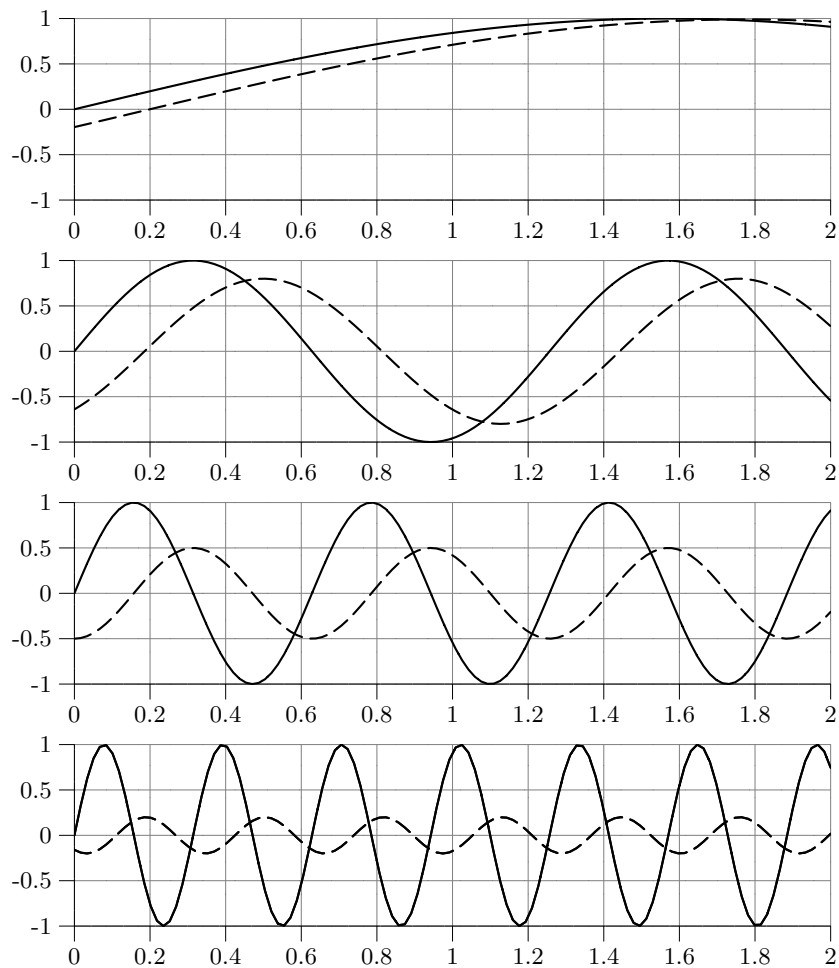


Figure 4.8a. $u(t) = \sin(\omega t)$ (solid) and $y(t)$ (dashed).

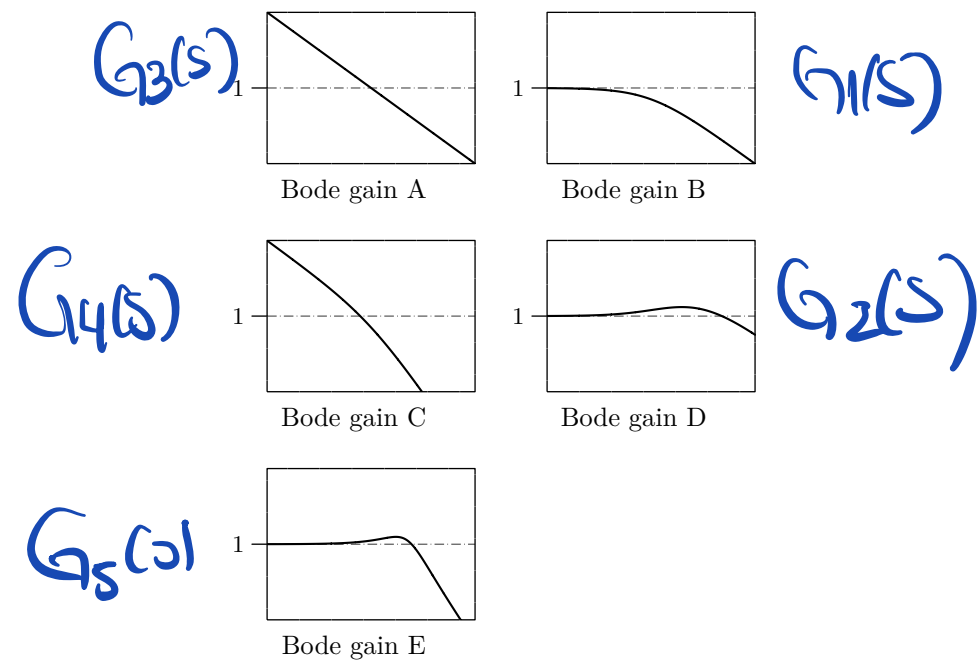


Figure 4.9a. All diagrams have equal scaling.

4.10 Figure 4.10a shows the Bode gain plots and step responses of four different systems, in no particular order. Identify the pair of plots that belongs to each system. That is, for each Bode gain plot (amplitude curve), find the corresponding step response. Motivate your answer by pointing out a set of unique features for each system.

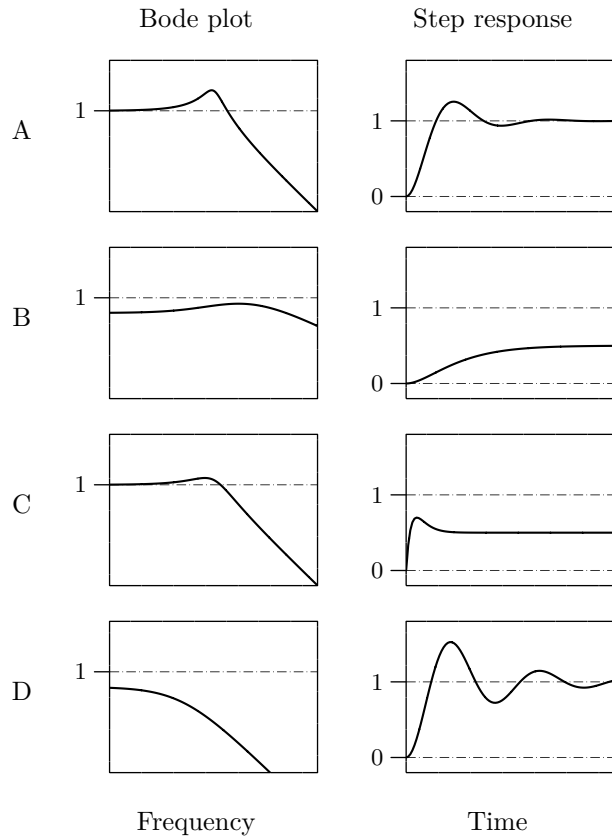


Figure 4.10a. All comparable diagrams have equal scaling.

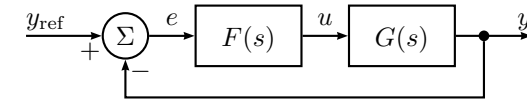


Figure 4.11a

4.11 The pH in a biochemical reactor is controlled by addition of a base. The transfer function $G(s)$ from added base to pH for the open system has been determined by experiments to be

$$G(s) = \frac{1.7}{(s+1)(0.7s+1)(0.5s+1)}$$

In an attempt to control the pH the control structure shown in Figure 4.11a is employed

- Make a Bode plot for the transfer function $G(s)$.
- Assume that a P controller is used ($F(s) = K$). At what value of K does the pH start to oscillate with constant amplitude?

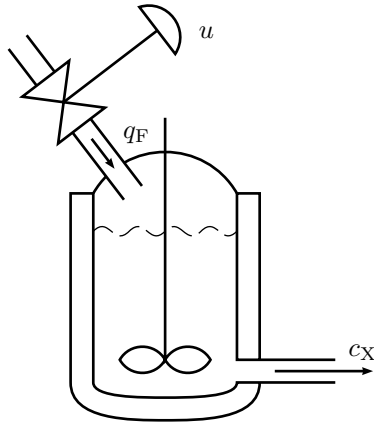


Figure 4.12a

4.12 Consider the biochemical reactor in Figure 4.12a. It is desirable to control the concentration of biochemical material c_X (output y) by manipulating the dilute flow q_F (input u). A model of this system can be described as

$$Y(s) = \frac{2e^{-5s}}{30s + 1}U(s)$$

where the time delay reflects the time it takes to measure the biochemical concentration. The bode diagram of the system is shown in Figure 4.12b

- For which values of K is a P controller going to stabilize the system?
- Construct a controller which has the crossover frequency $\omega_{c,d} = 0.1$ and no steady-state error.

4.13 A system

$$Y(s) = G_o(s)U(s)$$

is used with a controller

$$U(s) = R(s) - Y(s).$$

- In Figure 4.13a are Bode diagrams of the open loop system G_o for two different systems (top), the closed loop systems G_c (second row), step

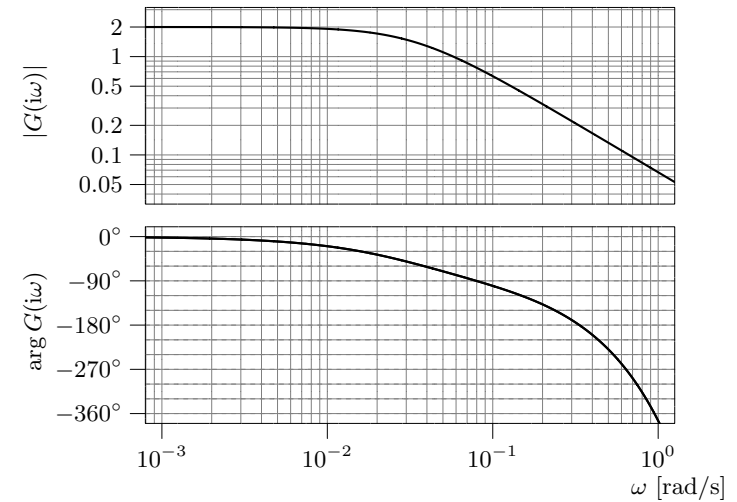


Figure 4.12b

responses of the closed loop systems (third row) and the poles of the closed loop systems (bottom). Assign each closed loop system, closed loop step response and closed loop pole diagram to the correct open loop system.

- Solve the same task for Figure 4.13b

4.14 **2010-10-19 Upg. 1a**

Då man applicerar insignalen

$$u(t) = \sin(2t), \quad t \geq 0$$

på ett första ordningens system

$$\dot{y}(t) + ay(t) = bu(t), \quad y(0) = y_0$$

får man utsignalen

$$y(t) = 2 \sin(2t - \pi/4).$$

Bestäm systemparametrarna a och b samt initialvärdet y_0 .

4.14 input

$$u(t) = \sin(2t) \quad t \geq 0$$

the first-order system

$$\dot{y}(t) + a y(t) = b u(t) \quad , y(0) = y_0$$

the output is

$$y(t) = 2 \sin(2t - \pi/4)$$

Determine a , b and y_0

① Transfer function

$$-y(0) + sY(s) + aY(s) = bU(s)$$

$$Y(s) = \frac{b}{(s+a)} U(s) + \frac{y(0)}{(s+a)}$$

← This dies after $t \gg 0$

Frequency response is in S.S.!!

$$s = j\omega$$

$$\frac{Y(s)}{U(s)} = \frac{b}{(s+a)} \xrightarrow{j\omega} \frac{b}{(j\omega + a)} = G(\omega)$$

$$|G(\omega)| = \frac{b^2}{\sqrt{\omega^2 + a^2}} \quad (1)$$

$$\angle G(\omega) = \text{atan}(0) - \text{atan}\left(\frac{\omega}{a}\right) = -\text{atan}\left(\frac{\omega}{a}\right) \quad (2)$$

We are given the info for $\omega = 2$ ($2t = \omega t$)

$$\text{Using (2)} = -a \tan\left(\frac{2}{a}\right) = -\frac{\pi}{4} \Rightarrow a = 2 //$$

Using (1)

$$\frac{b}{\sqrt{4+4}} \Rightarrow b = 4\sqrt{2} //$$

What about γ_0 ? Just use the equation

$$\gamma(t) = 2 \sin\left(2t - \frac{\pi}{4}\right) \Rightarrow 2 \sin\left(-\frac{\pi}{4}\right) = -\sqrt{2}.$$

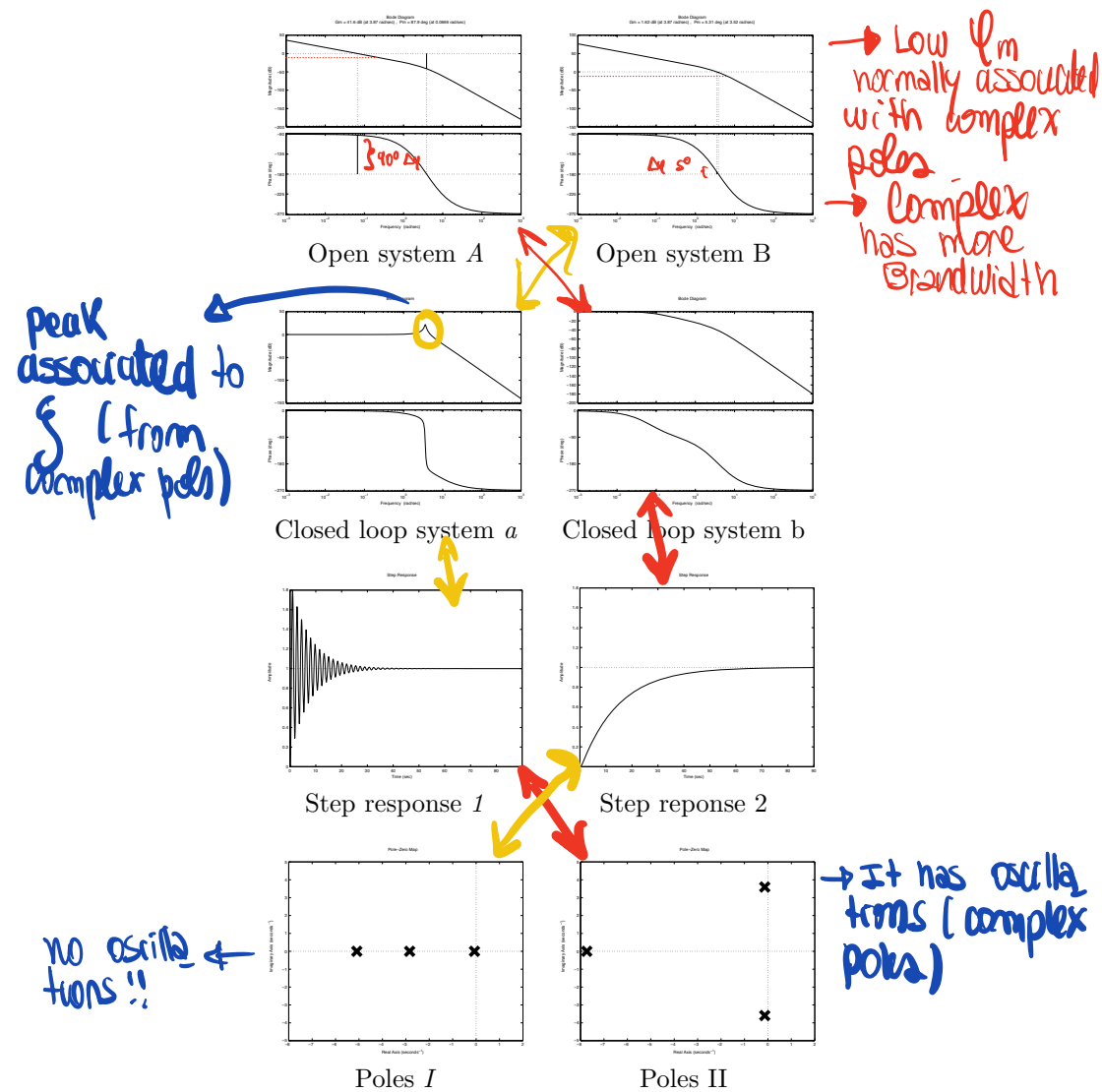


Figure 4.13a. The frequency scales of the Bode diagrams are the same. The time scales of the step responses are the same. In the pole-zero maps, $\times$ marks poles.

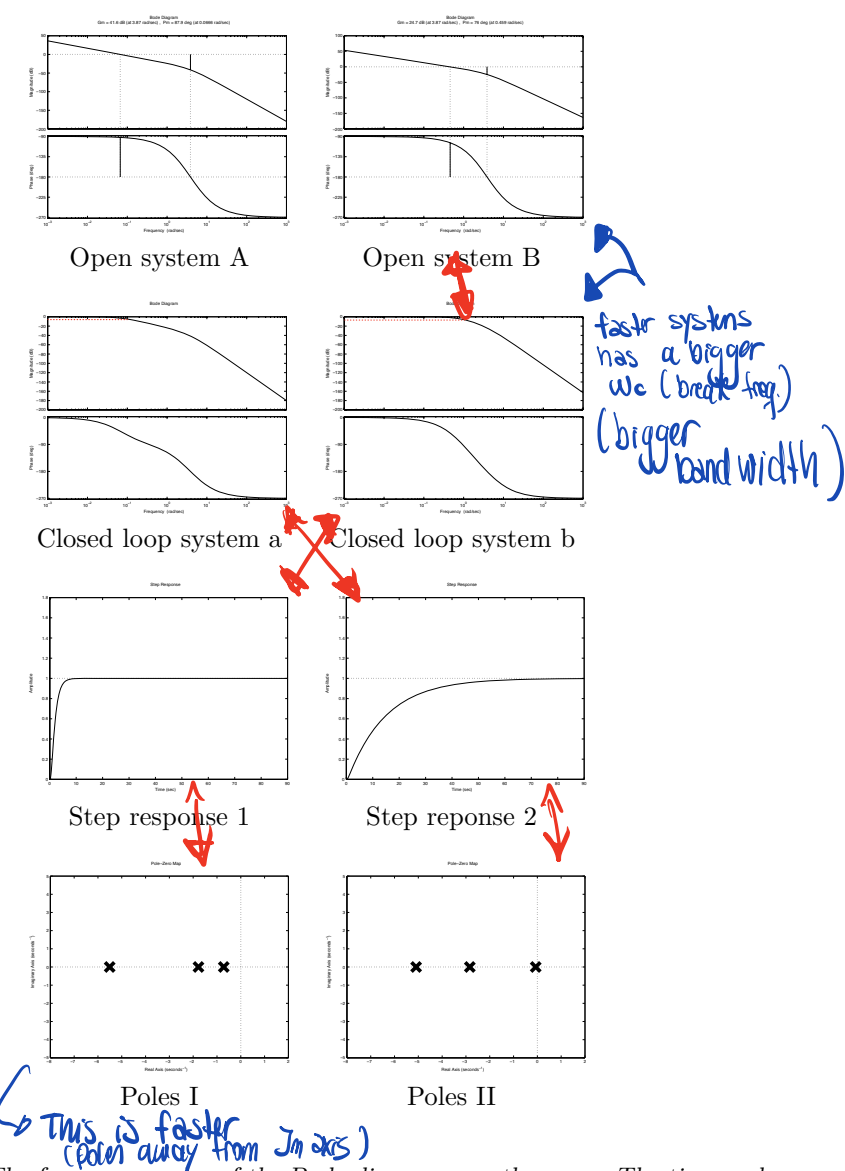


Figure 4.13b. The frequency scales of the Bode diagrams are the same. The time scales of the step responses are the same. In the pole-zero maps, $\times$ marks poles.